METRIC

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13 October 2026
SUPERSEDING
MIL-DTL-32632
25 July 2019

DETAIL SPECIFICATION

VALVES, FLOW CONTROL, DILATING DISK, FOR NAVAL SHIPBOARD WATER, OIL, AND GAS SERVICE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flow control valves of dilating, low torque, hysteresis-free design, employing multiple-segment irising seat-disks (petals) construction for naval shipboard water, oil (including fuel and JP-5 aviation fuel), and gas services. These valves are used for precise flow control of liquids and gases and allow for unconstrained flow when fully open.

1.2 Classification. Dilating disk valves (DDV) are classified by the material, fire rating, flange type, pressure-temperature rating, size, and operator to be used, as specified (see 6.2).

1.2.1 Types. DDVs are classified into three types, based on the materials of manufacture and fire resistance (see table I).

a. Type I – Non-ferrous
b. Type II – Corrosion-resistant steel (CRES)
c. Type III – CRES, fire resistant

1.2.2 End connections. DDV end connections are either socket weld or flanges. For socket dimensions and flange dimensions (outside diameter, thickness, drilling, etc.), see table III.

a. Welded
b. Navy flanged
c. ASME flanged

1.2.3 Valve and flange rating or class. See table III.

a. Navy 150
b. Navy 250
c. Navy 400
d. ASME 150
e. ASME 300
f. ASME 600
g. ASME 900
h. ASME 1500

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AMSC N/A FSC 4820

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1.3 Part or identifying number (PIN). The PIN to be used for parts acquired to this specification is constructed as follows:

M	32632	-	XXXX	X	X	-	XXX	X
Prefix for Military Specification	Specification Number	-	Classification (see 1.2 and code below)	Type (see 1.2.1)	End Type (see code below)	-	Size (see code below)	Operator (see code below)
Example: M32632-01503F-080H is for an ASME class 150, fire-resistant CRES, flanged end, Diamètre Nominal (DN) 80 (Nominal Pipe Size [NPS] 3), handwheel operated, dilating flow control valve.								

Classification	Code	Type	Code	End type	Code	NPS	Code (DN Size)	Operator	Code
Navy 150	N150	I (Non-ferrous)	1	Socket Welded	W	¼	008	Lever	L
Navy 250	N250	II (CRES)	2	ASME Flanged	F	1	025	Handwheel	H
Navy 400	N400	III (CRES, fire resistant)	3	Navy Flanged	N	1½	040	Electric Actuator	E
ASME 150	0150					2	050	Other	T
ASME 300	0300					3	080		
ASME 600	0600					4	100		
ASME 900	0900					6	150		
ASME 1500	1500					8	200		
						10	250		
						12	300		
						14	350		
						16	400		

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-5624 - Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)

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MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service

MIL-DTL-24657 - Valve Actuator, Direct Coupled, Gear-Driven Electro-Mechanical for Naval Shipboard Propulsion and Auxiliary Systems

MIL-DTL-46183 - Plastic Molding and Extrusion Material, Polyetheretherketone (PEEK)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-798 - Design Evaluation Requirements for Valves for Naval Shipboard Use

MIL-STD-1472 - Human Engineering

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVSEA DRAWINGS

803-1385620 - Handwheels for Valves

803-6983512 - Flanges, Standard, Cu-Ni Socket-Weld, Slip On, and Welding Neck

803-7063850 - Tailpiece Assemblies, Union End (BW, SW, SB), ¼-2 Inch, 400 PSI WOG

(Copies of these publications are online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required).)

NAVSEA PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). These publications are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this publication are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this publication are available online at www.aia-aerospace.org.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

ANSI/ISA 75.02 - Control Valve Capacity Test Procedures

(Copies of these publications are available online at https://webstore.ansi.org.)

AMERICAN PETROLEUM INSTITUTE (API)

API-STD-598 - Valve Inspection and Testing

API-STD-607 - Fire Test for Quarter-turn Valves and Valves Equipped with Non-Metallic Seats

(Copies of these publications are available online at https://www.apiwebstore.org/.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ Through NPS 24, Metric/Inch Standard

ASME B16.10 - Face-to-Face and End-to-End Dimensions of Valves

ASME B16.34 - Valves – Flanged, Threaded, and Welding End

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of these publications are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A108 - Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A276/A276M - Standard Specification for Stainless Steel Bars and Shapes

ASTM A351/A351M - Standard Specification for Castings, Austenitic, for Pressure-Containing Parts

ASTM A479/A479M - Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels

ASTM A564/A564M - Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

ASTM A743/A743M - Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Corrosion Resistant, for General Application

ASTM B16/B16M - Standard Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

ASTM B148 - Standard Specification for Aluminum-Bronze Sand Castings

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B164 - Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire

ASTM B446 - Standard Specification for Nickel-Chromium-Molybdenum-Niobium Alloy, Nickel-Chromium-Molybdenum-Silicon Alloy, and Nickel-Chromium-Molybdenum-Tungsten Alloy Rod and Bar

ASTM B505/B505M - Standard Specification for Copper Alloy Continuous Castings

ASTM D4894 - Standard Specification for Polytetrafluoroethylene (PTFE) Granular Molding and Ram Extrusion Materials

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ASTM F468 - Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

ASTM F593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs

ASTM F837 - Standard Specification for Stainless Steel Socket Head Cap Screws

(Copies of these publications are available online at www.astm.org.)

FLUID CONTROLS INSTITUTE, INC.

ANSI/FCI 70-2 - Control Valve Seat Leakage

(Copies of this publication are available online at www.fluidcontrolsinstitute.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 6708 - Pipework Components – Definition and Selection of DN (Nominal Size)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these publications are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AMS7259 - Rubber: Fluorocarbon (FKM), High Temperature/Fluid Resistant, Low Compression Set/85 to 95 Hardness for Seals in Fuel Systems and Specific Engine Oil Systems

SAE AMS7276 - Rubber: Fluorocarbon (FKM) High-Temperature-Fluid Resistant Low Compression Set/70 to 80 Type A Hardness for Seals in Fuel Systems and Specific Engine Oil Systems

SAE HS-1086 - Metals & Alloys in the Unified Numbering System

(Copies of these publications are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Materials. Unless otherwise specified (see 6.2), the materials for dilating flow control valves shall be as specified in table I. Use of other manufacturer recommended materials shall be authorized by the Technical Authority (see 6.6). Materials for components not specifically detailed herein shall be selected to prevent galling, seizing, and excessive wear of operating parts and galvanic compatibility with system piping.

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TABLE I. Materials.

Name of part	Type I (Nickel-copper [Ni-Cu] valves)		Type II and III (CRES valves)	
	Specification(s)	Material(s) 1/	Specification(s)	Material(s) 1/
Body and cover	MIL-B-24480, ASTM B148	UNS C95800	ASTM A182/A182M	Grade F304 (UNS S30400) or grade F316 (UNS S31600)
			ASTM A479/A479M	Type 304, UNS S30400
			ASTM A351/A351M	Grade CF8M (UNS J92900), CF3 (UNS J92500), or grade CF3M (UNS J92800)
			ASTM A743/A743M	Grade CF16F, UNS J92701 or UNS S30300
			ASTM A240/A240M	Grade F329, UNS S32900
Actuator stem adapter	ASTM A108	Steel, UNS G11170 or UNS G10200, nickel-plated	ASTM A108	Steel, UNS G11170 or UNS G10200, nickel-plated
Hex head cap screws or socket head cap screws	ASTM F468	Grade Ni 400	ASTM F593 or ASTM F837	CRES, UNS S30400
Body, bonnet, and stem seals	SAE AMS7276 or SAE AMS7259	FKM	SAE AMS7276 or SAE AMS7259	FKM
Interior seals	ASTM D4894	PTFE	ASTM D4894	PTFE
Actuator bracket	ASTM B150/B150M	316 CRES, UNS C63200	ASTM A276/A276M	304 or 316 CRES, UNS S30400 or S31600
	MIL-B-24480	UNS C95800		
Stem, pivot pins, arms, ring gear	ASTM B164	Ni-Cu, UNS N04400	ASTM A564/A564M	Steel, UNS S17400, H1100, heat treated

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TABLE I. Materials – Continued.

Name of part	Type I (Nickel-copper [Ni-Cu] valves)		Type II and III (CRES valves)	
	Specification(s)	**Material(s)** [1/]	**Specification(s)**	**Material(s)** [1/]
Petals and seals [2/]	ASTM B164 petals with ASTM D4894 seals	Ni-Cu, UNS N04400 petals with PTFE seals [3/]	ASTM A240/A240M petals with ASTM D4894 seals	UNS S31600 petals with PTFE seals [2/]
	ASTM B446 petals with ASTM D4894 seals	Inconel 625 (UNS N06625) or similar petals with PTFE seals [3/]		
Pinion bearing	ASTM B505/B505M	Nickel-aluminum bronze, UNS C95500	ASTM B16/B16M	Brass, UNS C36000, cold-drawn H02
Stem base bushing	MIL-DTL-46183, type III, class 2	Carbon-filled PEEK	MIL-DTL-46183, type III, class 2	Carbon-filled PEEK
Cover ring and body ring bearings, stem bushing	ASTM D4894	PTFE	ASTM D4894	PTFE
Bonnet, cover and body	ASTM B61	UNS C92200	ASTM A351/A351M	UNS J92900

NOTES:

[1/] Table I contains the commercial and military specifications and standards and, in addition, contains the corresponding SAE HS-1086 Unified Numbering System (UNS) designation.

[2/] Due to performance characteristics or working conditions, other seal or petal seal materials may be used with prior Technical Authority approval (see 6.6).

[3/] Soft seat material (PTFE) may not be necessary for metal-to-metal seated valves.

3.2.1 Prohibited materials. Valves shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction. Dilating flow control valves shall consist of three or more rotationally closing plate segments, positioned by gearing. Gearing may be ring, or rack, and pinion. The operating stem shall not enter the flow passage. The valves shall be tight seating at all ports. Valve bodies shall withstand the applied stresses developed during operation of the valve. Valves shall be operable by one person in accordance with MIL-STD-1472. Figure 1 illustrates a typical valve of this construction.

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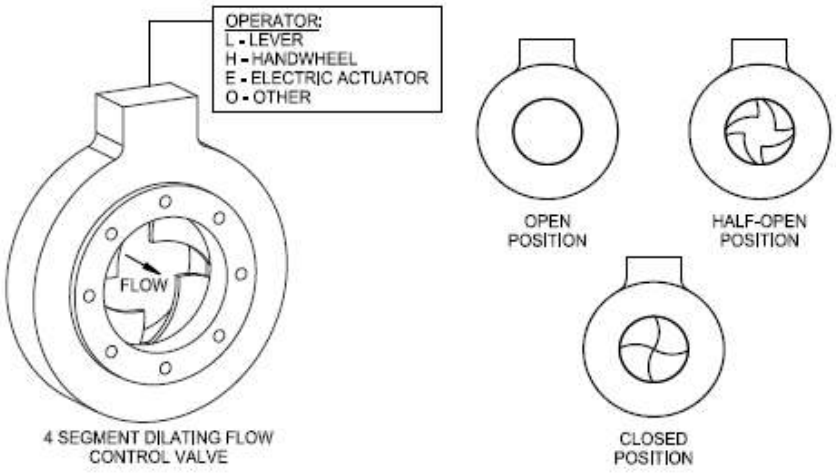


FIGURE 1. Typical dilating flow control valve.

3.3.1 Dimensions. Unless otherwise specified (see 6.2), the face-to-face, center-to-face, and end-to-end dimensions for flanged valves shall be in accordance with the ASME B16.10 and ASME B16.34 requirements for long pattern, steel globe valves, for the valve size specified. Flow passage shall be round bore. Dimensions and machining of end flanges and sockets shall be in accordance with table III. Valve flange bolt holes may be tapped to suit bolting in areas where valve body interferes with through bolting.

3.3.2 Petal (segment) motion. Dilating flow control petal (segment) motion shall be limited by positive stops at the full-open and full-closed positions. The valve segment motion shall have complete throttling capability between the full-open and full-closed positions. The position shall be shown by permanent and distinct external marks on top of the operating stem. The valve shall incorporate provisions for positively retaining the dilating disk assembly in all positions.

3.3.3 Low friction surfaces. Valve components that rotate shall operate against a low friction seat or seal face material. Low friction treated surfaces shall be either coated or sleeved in accordance with ASTM D4894. Surfaces that rub on one another shall be designed to reduce galling.

3.3.4 Handwheels and lever handles. Handwheels smaller than 279 millimeters (mm) (11 inches) in diameter and all levers shall be of commercial design and manufactured from steel, ductile iron, malleable iron, or cast manganese bronzes. Handwheels that are 279 mm in diameter and larger shall be cast steel in accordance with the general dimensions of Drawing 803-1385620. Aluminum or aluminum alloys shall not be used for handwheels or levers. The stem handwheel connection shall be in accordance with Drawing 803-1385620 for all sizes.

3.3.5 Handwheel and lever handle size. Unless otherwise specified (see 6.2), valves under DN150 (6 inches) shall be furnished either with lever handles or handwheels. Valves that are DN150and larger shall be provided with handwheels operated through enclosed worm and gear mechanisms. Handwheels shall be not less than 152 mm in diameter. Operating forces, nominal lever lengths, and nominal handwheel diameters shall be as specified in table II.

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TABLE II. Valve operating torque.

Valve nominal size DN [1/] (NPS)	Maximum allowable tangential force [2/] Newtons (pounds-force)	Maximum operating torque [3/] Newton-Meters (foot-pounds)	Nominal lever length [4/] mm (inches)	Nominal handwheel diameter [4/] mm (inches)
8 (¼) and 10 (½)	54.2 (40)	16.3 (12)	127 (5)	178 (7)
25 (1)	54.2 (40)	17.6 (13)	152 (6)	203 (8)
40 (1½)	54.2 (40)	20.3 (15)	178 (7)	230 (9)
50 (2)	54.2 (40)	24.4 (18)	178 (7)	279 (11)
80 (3)	54.2 (40)	27.1 (20)	229 (9)	305 (12)
100 (4)	54.2 (40)	36.6 (27)	279 (11)	406 (16)
150 (6)	54.2 (40)	47.5 (35)	--	533 (21)
200 (8)	54.2 (40)	54.2 (40)	--	610 (24)
250 through 400 (10 through 16)	54.2 (40)	54.2 (40)	--	Up to 914 (36)

NOTES:

1/ Sizes in accordance with ISO 6708.

2/ Measured at the rim of the handwheel.

3/ Measured at valve stem for hand-operated valves and measured at manual-actuator stem for valves with gear boxes.

4/ Nominal lever length and handwheel diameters are provided for torque measurement requirements and are not product requirements.

3.3.6 Manual valve actuators. Worm and gear mechanisms shall be enclosed and shall be commercial items, except that they shall be shock tested (see 4.6.1) and vibration tested (see 4.6.2) while mounted on the valve in an operating condition.

3.3.7 Remote actuating gear. Valves shall be capable of use with mechanical remote actuating gear. The adapter for remote actuating gear shall be as specified (see 6.2). Valves shall accept remote actuating gear sized to deliver between 20 and 50 percent more torque than the maximum seat torque listed in table II.

3.3.8 Electric valve actuators. When specified (see 6.2), valves shall be capable of use with electrically-powered valve actuators in accordance with MIL-DTL-24657 or commercial equivalents. DOD-V-24657 actuators may be used with prior Technical Authority approval (see 6.6). The actuators shall be shock tested (see 4.6.1) and vibration tested (see 4.6.2) while mounted on the valve in an operating condition.

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3.4 Performance.

3.4.1 Pressure and temperature rating. Pressure and temperature rating shall be as specified in table III.

TABLE III. Service rating and end connections.

Type	Nominal size DN [1/] (NPS)	End connection		Service rating		
		Type	Dimensions and machining	Classification	Pressure (psi)	Temperature °C (°F)
I	8 (¼) through 50 (2)	Union end, butt, or socket weld tailpieces	803-7063850	Navy 150 Navy 250 Navy 400	150/100[3/] 250/150[3/] 400/200[3/]	66/218 (150/425) [3/]
	10 (1/2) through 400 (16)	Flanged, flat face	803-6983512 [2/]			
II & III	50 (2) and below	Socket weld	ASME B16.11 [4/]	ASME 150 ASME 300 ASME 600 ASME 900 ASME 1500	Class 150 [5/] Class 300 [5/] Class 600 [5/] Class 900 [5/] Class 1500 [5/]	
	8 through 400 (¼ through 16)	Flanged, raised face	ASME B16.5			

NOTES:

1/ Sizes in accordance with ISO 6708.

2/ Valve flange dimensions shall be in accordance with Drawing 803-6983512 for flanges whose dimensions are in accordance with MIL-PRF-20042 and have the same service rating as specified herein.

3/ Ratings between the values provided can be determined by linear interpolation.

4/ For socket bore diameter and minimum depth.

5/ In accordance with the ASME B16.34 requirements for standard class.

3.4.2 Working fluid. Dilating flow control valves shall normally serve in water, oil, JP-5 (MIL-DTL-5624), or gas services. When specified (see 6.2), valves shall serve seawater, sewage, or other working fluids containing high contaminant or solids loads and meet requirements herein.

3.4.3 Shock. Unless otherwise specified (see 6.2), valves shall withstand the grade A, hull-mounted, class I, type C shock requirements in accordance with MIL-DTL-901 and MIL-STD-798 (see 4.6.1).

3.4.4 Vibration. Unless otherwise specified (see 6.2), valves shall withstand the type I vibration requirements in accordance with MIL-STD-167-1 (see 4.6.2).

3.4.5 Leak through. When closed, valves through leakage shall meet the class V criteria of ANSI/FCI 70-2 or the criteria of API-STD-598, as specified (see 6.2).

3.4.6 Hydrostatic withstand. Valves shall withstand an internal hydrostatic test (see 4.6.5) at the pressure specified by ASME B16.34 for its rating, in the open position, for 5 minutes with no external leakage, weeping, or structural distortion.

3.4.7 Maintainability. Valve maintenance shall be accomplishable without the need for special training or special tools. Special tools are defined as those tools not listed in the GSA Global Supply Catalog. Valves shall

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operate without maintenance requirements, except operating mechanism lubrication, for 20,000 open-close operating cycles.

3.4.8 Fire resistance. When specified (see 6.2), dilating disk valves shall be fire resistant and shall meet the allowable through-valve and external leakage rate acceptance criteria when tested in accordance with API-STD-607 (see 4.5.3).

3.5 Marking.

3.5.1 Body marking. Valve bodies shall have the pressure rating and manufacturer's name or trademark etched, stamped, or cast integral with the body.

3.5.2 Identification plates. Valves shall bear an identification plate made of CRES or brass. Identification plates shall be permanently fastened to a part of the valve not subjected to the working pressure. The identification plate shall include the following information:

a. Manufacturer's name or trademark.

b. Size of valve and pressure rating.

c. Material composition of body and petal segments.

d. Manual valve actuator mechanical advantage (if applicable).

e. Manufacturer's identification number or drawing number.

f. Specification number "MIL-DTL-32632".

g. PIN (see 1.3).

3.6 Workmanship. Valves shall be thoroughly cleaned both inside and out. Valves shall be free from burrs, loose sand, grit, scale, metal shavings, and all other contaminants that can affect performance and damage sealing surfaces. There shall be no sharp edges. All sealing, rubbing, and coated surfaces shall be smooth with no blemishes, scratches, scrapes, or gouges. Machined surfaces shall not exceed 125 roughness height rating as defined in ASME B46.1, unless a rougher finish is allowed by specifications. Valve bodies shall meet the visual acceptance criteria for castings in accordance with S9074-AR-GIB-010/278.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one valve of each class when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5 and the tests specified in table IV. Acceptance of the inspected sizes shall constitute acceptance of the sizes not inspected as specified in table V.

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TABLE IV. First article and conformance inspection.

Characteristic	Requirement	First article inspection	Conformance inspection
Materials	3.2	4.5.1	--
Visual and dimensional examination	3.3, 3.4.1, 3.4.2	4.5.2	4.5.2
Examination after first article tests	3.3	4.5.3	--
Petal movement and throttling	3.3.2	4.6.4.2	--
Shock	3.4.3	4.6.1	--
Vibration	3.4.4	4.6.2	--
Leak through	3.4.5	4.6.3	--
Operational	3.4.6	4.6.4, 4.6.5	--
Hydrostatic shell and functional	3.4.6	4.6.5	4.6.5
Maintainability	3.4.7	4.5.3	--
Markings	3.5	4.5.2	4.5.2
Workmanship	3.6	4.5.2	4.5.2
Fire resistance	3.4.8	4.8	--

TABLE V. First article approval sizes.

Size tested DN (NPS)	Approves these sizes DN (NPS)
8 (¼)	8 (¼)
50 (2)	10 (½), 25 (1), 40 (1½), 50 (2)
150 (6)	80 (3), 100 (4), 150 (6)
250 (10)	200 (8), 250 (10)
300 (12)	300 (12)
350 (14)	350 (14)
400 (16)	400 (16)

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests specified in table IV. Every valve offered to the Government shall be tested in accordance with 4.6.5. Conformance inspection shall be performed on samples selected in accordance with 4.4 or as specified (see 6.2).

4.4 Sampling.

4.4.1 Sampling for examinations and tests. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.4.2 Lot. A lot shall consist of all valves of the same class and size produced under the same conditions and offered for delivery at one time, as specified. A sample valve shall consist of one completely assembled dilating flow control valve.

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4.5 Examinations.

4.5.1 Examination of materials. Materials used in the manufacture of valves shall be examined to determine conformance to 3.2, as applicable (see 6.2).

4.5.2 Visual and dimensional examination. Sample valves selected shall be visually and dimensionally examined to verify compliance with the nonoperational requirements of this specification and the manufacturer's drawings. Each of the sample valves selected shall be examined in accordance with table IV and table VI. Any valve in the sample containing one or more defects shall be rejected, and if the number of defective valves in any sample exceeds the acceptable number for that sample, the lot represented by the sample shall be rejected.

TABLE VI. Classification of defects.

Categories	Major defects
101	Class not as specified (see 1.2)
102	Material not as specified (see 3.2)
103	Construction not as specified (see 3.3.1)
104	Shock resistance not as specified (see 3.4.3)
105	Vibration resistance not as specified (see 3.4.4)
106	Leak through not as specified (see 3.4.5)
107	External leakage not as specified (see 3.4.6)
108	Maintainability not as specified (see 3.4.7)
110	Body marking and identification plates not as specified (see 3.5)
111	Workmanship not as specified (see 3.6)

4.5.3 Examination after first article tests. Upon conclusion of first article tests, the valve shall be disassembled and visually and dimensionally examined for damage and wear. Any valve or sample lot not meeting defect criteria of table VI shall be rejected. Disassembly and reassembly shall be performed to determine the practicability of maintaining a valve of this design for shipboard use. A maintainability demonstration shall be conducted by removing and replacing elastomers, petal segments, and operating gear. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used. Particular attention shall be given to avoid the possible loss of small parts by the maintenance crew.

4.6 Tests.

4.6.1 Shock. The valve shall be subjected to the grade A shock test in accordance with MIL-DTL-901 to determine conformance to 3.4.3. Valves shall be hydrostatically pressurized internally to the maximum gauge pressure as specified in 3.4.1. The testing medium shall be clean tap water at temperatures not less than 4 °C (40 °F) or exceeding 52 °C (125 °F). The medium may not contain additives, except for corrosion inhibitors that do not increase the viscosity or surface tension of the water and residual cutting oil from machining. Momentary malfunction at the time of impact blow is permitted and acceptable. Permanent deformation, misalignment, or functional impairments shall be cause for rejection. Permissible seat leakage shall be as specified in 3.4.5.

4.6.2 Vibration. The valve shall be subjected to the type I vibration test in accordance with MIL-STD-167-1 to determine conformance to 3.4.4.

4.6.3 Leak through and seat leakage. The valve shall be tested in accordance with specified API-STD-598 or ANSI/FCI 70-2 criteria or better (see 3.4.5 and 6.2).

4.6.4 Operational tests. Operational tests shall be performed on each valve test size and class in accordance with table V. Tests shall alternately apply the pressure specified in 3.4.1 with the pressure on the seat with the valve

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in the shut position for 20,000 cycles. One cycle shall consist of the pressure specified in 3.4.1 applied in the shut position. At the completion of the operational test, the external leakage test of 4.6.5.2 shall be performed.

4.6.4.1 Operating torque test following a period of non-operation. A differential pressure equal to the rated pressure shall be maintained across the valve for 40 hours. After this period of time has elapsed, the valve shall be fully cycled (closed to open to closed) every 2 hours and the breakaway torque recorded. This shall be continued until such time as three consecutive readings that are both within ±5 percent of each other are obtained. Any torque measurement greater than the maximum operating torque (see table II) shall be cause for rejection.

4.6.4.2 Throttling test. Each test valve size and class shall be tested to determine conformance with 3.4.5, set in the 15 to 17 percent open position, and clean tap water shall be passed through the valve for a minimum of 12 hours. Upon completion of this test, the valve shall be tested for leakage in both the flow directions using clean, ambient temperature tap water at the 38 °C (100 °F) rated pressure. The test pressure shall be held for a minimum of 1 minute. The water leakage rate shall not exceed the maximum seat leakage rate of 4.6.3.

4.6.5 Hydrostatic shell pressure, external leakage, and functional check. Every valve offered to the Government shall be tested in accordance with 4.6.5.1 and 4.6.5.2 after being fully cycled (opened and closed) three times to demonstrate acceptable assembly and operation. When actuators are to be used, this test shall be conducted with actuators installed.

4.6.5.1 Test conditions. Unless otherwise specified (see 6.2), each valve shall be hydrostatically tested for structural integrity and seat leakage using clean tap water. In cases where the expected working fluid is seawater, sewage, or another fluid that contains high levels of contaminants or solids loads, the test fluid shall be the expected working fluid. Tests shall be conducted within a temperature range of 4° to 52 °C (40° to 125°F) and the media shall contain no additives other than cutting oil. Test fixtures shall be adjusted to permit expansion and elongation of the valve body due to internal pressure to the fullest extent possible. The thrust on the ends shall be the minimum required to facilitate testing and prevent leakage to the atmosphere.

4.6.5.2 External leakage hydrostatic test. Each valve shall be subjected to 150 percent of the rated pressure in the open position for 5 minutes. Any external weeping or leakage during or after pressurizing or permanent deformation after pressurizing shall be cause for rejection.

4.7 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.1 and 6.2).

4.8 Fire test. When specified (see 3.4.8 and 6.2), valves shall be fire tested as follows:

4.8.1 Procedure. Valves shall be fire tested in accordance with API-STD-607. Test sizes shall be selected from the “Other Valves Qualified by NPS” table of API-STD-607. When selecting test sizes, consideration should be given to selecting the least number of valve sizes for fire testing to encompass a range of sizes of the same basic design that require qualification. Leakage shall be measured by monitoring the source tank water levels during the test.

4.8.2 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority (see 6.6). Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.a2la.org) or the International Code Council’s International Accreditation Services (IAS) (www.iasonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Valves covered by this specification are high-reliability, are intended for service aboard Navy ships with a 30-year service life, and meet Navy shock and other unique requirements. Existing stock of dilating disk valves manufactured to MIL-DTL-32632, dated 25 July 2019, are acceptable for the uses they are suited to, until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Valve classification, end type, size, and operator to be used (see 1.2).

c. When first article inspection is required (see 3.1 and 6.3).

d. Materials, if other than specified, as authorized by the Technical Authority (see 3.2 and 6.6).

e. Requirement for prohibited materials verification (see 3.2.1 and 4.7).

f. Valve face-to-face, center-to-face, and end-to-end dimensions, if other than specified (see 3.3.1).

g. Lever handle or handwheel requirements, if other than specified (see 3.3.5).

h. Remote actuating gear adapter requirements (see 3.3.7).

i. If valves are required to be capable of use with electric valve actuators (see 3.3.8).

j. Expected working fluid, if containing high contaminant or solids loads, such as seawater or sewage (see 3.4.2).

k. Shock requirements, if other than specified (see 3.4.3).

l. Vibration requirements, if other than specified (see 3.4.4).

m. Applicable leak through requirements, either API-STD-598 or ANSI/FCI 70-2 (see 3.4.5 and 4.6.3).

n. Fire resistance and testing, if required (see 3.4.8 and 4.8).

o. Additional conformance inspections, if required (see 4.3).

p. AQL of 1.0 (see 4.4.1).

q. Specific materials testing requirements, if required (see 4.5.1).

r. Test fluid, if not clean tap water (see 4.6.5.1).

s. Additional fire testing provisions (see 4.8.2 and 6.5).

t Packaging requirements (see 5.1).

u. Documentation, if required (see 6.4).

6.3 First article. When first article inspection is required (see 6.2), the contracting officer should provide specific guidance to offerors that the item is a first article sample (see 3.1) and the number of items to be tested as specified in 4.4. Acquisition documents should include specific instruction regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitation.

6.4 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any valve under this specification:

a. Engineering drawings.

b. Materials listing.

c. Installation drawings.

d. Technical manual.

e. Test reports.

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f. Certifications.

g. Onboard repair parts list.

6.5 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.a2la.org) or International Code Council's IAS (www.iasonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.6 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05T may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.7 Subject term (key word) listing.

Iris
Petal
Segment
Shutter

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 69

Preparing activity:
Navy – SH
(Project 4820-2026-011)

Review activities:
Army – MI
Navy – AS
Air Force – 03, 71
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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